

PageRank-Based Vaccination Strategies

Open Ended Project - Networks and Markets

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1 Introduction

The rapid spread of epidemics in networks, such as infectious diseases in human populations or computer viruses in digital networks, poses significant threats [?, ?]. Controlling such epidemics requires effective immunization strategies that can minimize the spread by vaccinating a subset of nodes [?, ?], for instance in the case of short supply of vaccines or limited resources, such as at the beginning of a pandemic—as we have seen with COVID-19. Traditional immunization approaches often focus on preemptive vaccination without considering the current state of the epidemic, which may not be optimal once an outbreak has begun.

In this work, we address the *Data-Aware Vaccination (DAV)* problem, which involves selecting the best set of nodes to vaccinate, given prior information about the infection’s spread [?, ?]. Without knowledge of existing DAV algorithms, we independently hypothesized that **PageRank centrality** could serve as an effective heuristic for identifying critical nodes for vaccination. PageRank is a measure of node importance based on the structure of the network and has been widely used in ranking web pages [13].

We compare the performance of various vaccination strategies, including Random, Degree-based, PageRank-based, and NetShield algorithms [17], under the Independent Cascade (IC) and Susceptible-Infected-Recovered (SIR) models [5, 6]. Our experiments on real-world networks demonstrate that the PageRank-based strategy significantly outperforms other heuristics in minimizing the epidemic’s spread.

Contributions:

- We propose using PageRank centrality as a heuristic for node vaccination in the DAV problem.
- We implement and evaluate the effectiveness of various vaccination strategies on real-world networks.
- We demonstrate that the PageRank-based strategy significantly reduces the spread of epidemics compared to other heuristics.

2 Related Work

Vaccination strategies in networks have been extensively studied [?, ?]. Preemptive immunization methods often rely on network centrality measures, such as degree or betweenness, to select nodes for vaccination [?]. Data-aware approaches consider the current state of the epidemic to make vaccination decisions [?, ?].

The NetShield algorithm [17] aims to select nodes that minimize the spectral radius of the adjacency matrix, thereby reducing the network’s vulnerability to epidemic spread. Our work differs in that we explore the use of PageRank centrality as a heuristic for data-aware vaccination without prior knowledge of such algorithms.

3 Background

Let $G=(V,E)$ be an undirected graph representing the network, where V is the set of vertices and E is the set of edges. Let $S \subseteq V$ represent the set of initially infected nodes. The objective is to select a subset $T \subseteq V \setminus S$ of k healthy nodes to vaccinate, minimizing the spread of the epidemic.

This allows us to reason both about the spread of diseases in human populations, where nodes represent individuals, and about the spread of computer viruses in digital networks, where nodes represent computers or servers.

3.1 Epidemic Models

We consider two epidemic models, each suited to different disease dynamics:

Susceptible-Infected-Recovered (SIR) Model [6] The SIR model is commonly used to study disease spread in networks [1, 4, 16]. It models infectious diseases with varying recovery rates, and we thus use it to model the low-recovery rate disease like Ebola, known for its high mortality. Specifically, infected nodes can transmit the disease to susceptible neighbors with a high probability $p=0.8$, and recover with a low probability $\delta=0.1$, becoming immune and ceasing transmission.

Independent Cascade (IC) Model [5] A variant of the SIR model, the IC model is used to simulate information diffusion in networks [5]—by allowing nodes to infect neighbors only once. We use it to model diseases like COVID-19, where quick quarantine and recovery are common. Specifically, infected nodes infect susceptible neighbors with probability $p=0.4$, but can only attempt infection once before becoming inactive.

3.2 PageRank Centrality

PageRank is a measure of node importance based on the idea that connections from important nodes contribute more to the importance of a node [13]. The PageRank score $PR(i)$ of node i is defined recursively as:

$$PR(v) = \frac{\epsilon}{N} + (1-\epsilon) \sum_{(v',v) \in E} \frac{PR(v')}{\text{out-degree}(v')}$$

where ϵ is the damping factor (typically set to $1/7$), and N is the total number of nodes.

4 Problem Formulation

The **Data-Aware Vaccination (DAV)** problem involves selecting a set $S^* \subset V \setminus S$ of k healthy nodes to vaccinate to minimize the expected number of infected nodes $\sigma_{G,S}(S)$ at the end of the epidemic:

$$S^* = \arg \min_{S \subset V \setminus S, |S|=k} \sigma_{G,S}(S).$$

This problem is known to be NP-hard [?], and exact solutions are computationally infeasible for large networks. Therefore, heuristic approaches are necessary.

5 Methodology

5.1 Vaccination Strategies

We implement and compare the following strategies:

Strategy	Description
No Vaccination	No nodes are vaccinated.
Random	Randomly select k healthy nodes to vaccinate.
NetShield	Use the NetShield algorithm [17] to select nodes that minimize the spectral radius.
Degree-based	Select the k healthy nodes with the highest degrees. A degenerate case of PageRank, as discussed in class.
PageRank-based	Select the k healthy nodes with the highest PageRank scores.

5.2 Experimental Setup

Social Networks We conduct experiments on four real-world social and peer-to-peer networks, detailed in [Appendix B](#).

Epidemic Propagation Models Configurations We simulate both the IC and SIR models on each network, to model different disease dynamics, discussed in [Section 3.1](#). The configuration parameter ranges are varied based on the network size. We vary the vaccination budget size to evaluate the strategies' performance under different resource constraints.

Network	Initial Infection Size ($ S $)	Vaccination Budget (k)	Simulations Per Configuration
Facebook, LastFM, Gnutella	100	10,30,50,70,90,110,130,150	200
Brightkite	400	10,50,100,150,200,250,300,350	50

Evaluation Metric We evaluate each strategy based on the **average number of healthy nodes** remaining after the epidemic. A higher number of healthy nodes indicates a more effective vaccination strategy. This is averaged across the amount of simulations per configuration.

6 Experimental Results

Due to space limitations, we present a summary of our experimental results here. Detailed figures for each network under both epidemic models are provided in the [Artifacts Appendix](#) (see Figures 1 and 2).

Our experiments consistently demonstrate that the PageRank-based vaccination strategy significantly reduces the spread of epidemics compared to other heuristics across different networks and epidemic models. Moreover, we note a significant runtime improvement compared to the NetShield algorithm, which requires solving an optimization problem to minimize the spectral radius.

7 Discussion

The effectiveness of the PageRank-based strategy can be attributed to its ability to identify nodes that are not only highly connected but also central in the network's structure. These nodes have a higher potential to influence a large portion of the network [?].

Our findings align with the principles of data-aware vaccination, which emphasizes the importance of utilizing network information and the current state of the epidemic in vaccination decisions [?]. While NetShield is designed to minimize the spectral radius, our results suggest that PageRank centrality offers a more practical and computationally efficient heuristic.

8 Conclusion

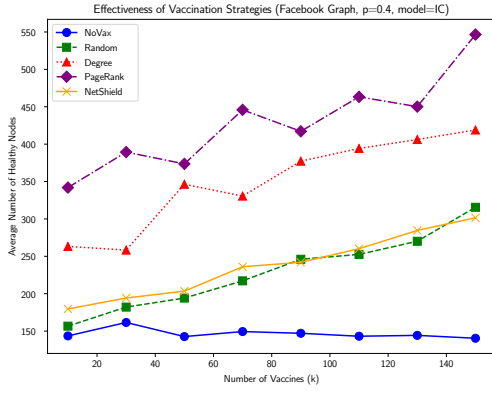
We have shown that using PageRank centrality as a heuristic for node vaccination in the Data-Aware Vaccination problem is highly effective in controlling the spread of epidemics in networks. Our extensive experiments on real-world networks and epidemic models validate our hypothesis and demonstrate that the PageRank-based strategy outperforms other heuristics.

Future work includes exploring dynamic vaccination strategies that adapt to the evolving state of the epidemic and investigating the combination of PageRank with other centrality measures to further enhance vaccination effectiveness.

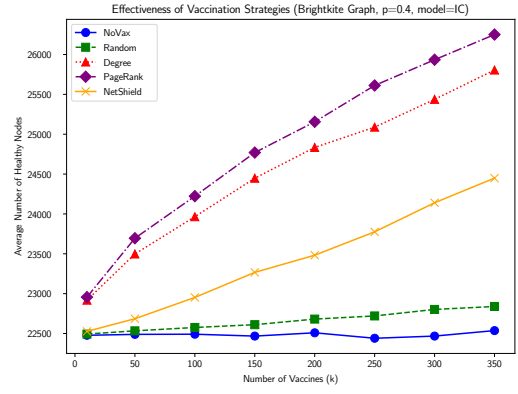
A Artifacts

In this appendix, we present the experimental results in the form of two composite figures, each containing four subfigures corresponding to the different networks analyzed under each epidemic model.

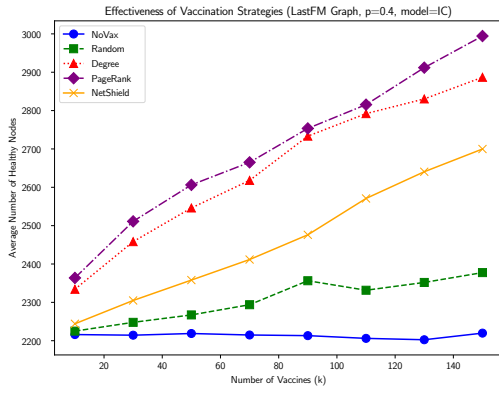
A.1 Independent Cascade Model Results (COVID-19-like)



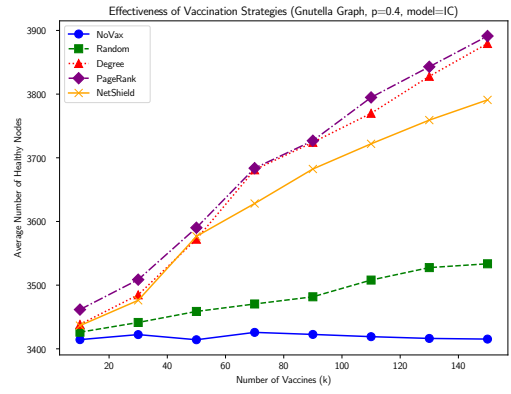
(a) Facebook Network



(b) Brightkite Network



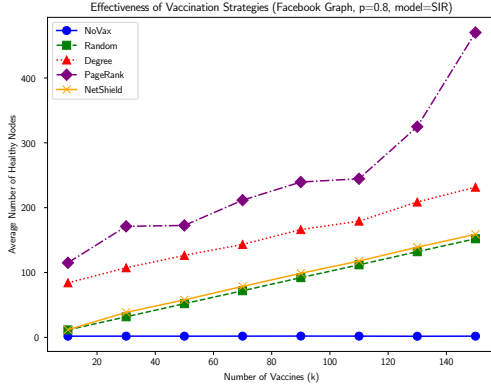
(c) LastFM Network



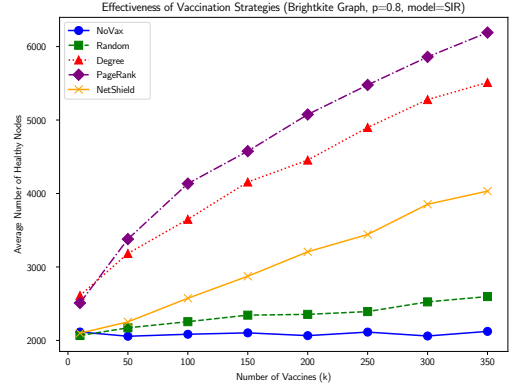
(d) Gnutella Network

Figure 1: Effectiveness of Vaccination Strategies under the Independent Cascade Model ($p=0.4$)

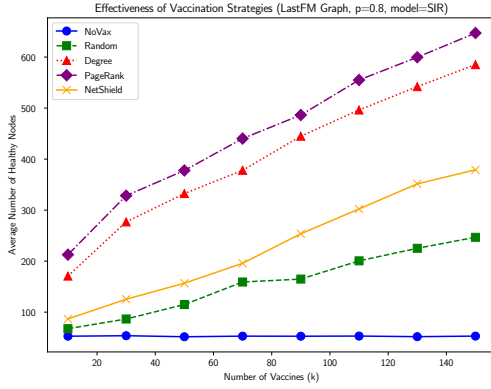
A.2 SIR Model Results (Ebola-like)



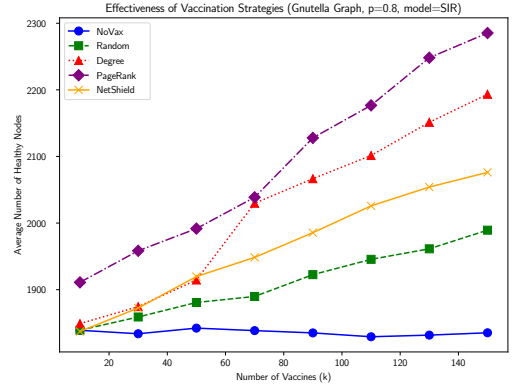
(a) Facebook Network



(b) Brightkite Network



(c) LastFM Network



(d) Gnutella Network

Figure 2: Effectiveness of Vaccination Strategies under the SIR Model ($p=0.8$, $\delta=0.1$)

B Datasets

We used the following real-world network datasets in our experiments, taken from the [8, Stanford Network Analysis Project (SNAP)]:

Dataset	Description	Nodes	Edges
Facebook Social Network [3, 9]	A network representing friendships on Facebook.	4,039	88,234
Brightkite Location-Based Social Network [11, 2]	A network from the Brightkite location-based social networking platform. We used the friendship network for our experiments.	58,228	214,078
LastFM Music Social Network [12, 15]	A network representing friendships on LastFM, a music recommendation platform.	7,624	27,806
Gnutella Peer-to-Peer Network [10, 7, 14]	A network of Gnutella peer-to-peer file sharing connections. It serves as a model for online virus spread, expanding the scope of our experiments.	6,301	20,777

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